

Smart Fire Detection using ESP32 and Blynk IoT

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Abstract — Wildfires cause significant environmental, economic, and human losses worldwide. Traditional wildfire detection methods depend on human observation, watchtowers, and emergency calls, which often result in delayed response times. The system uses multiple sensors including temperature-humidity sensors, gas sensors, and infrared flame sensors to detect abnormal environmental conditions. Sensor fusion techniques are employed to reduce false alarms and improve detection accuracy. The collected data is processed by an ESP32 microcontroller and transmitted through LoRaWAN communication technology to cloud platforms. GPS modules provide precise location coordinates of detected fire incidents. Solar-powered sensor nodes ensure long-term autonomous operation in remote forest regions. Cloud services analyze incoming data and generate alerts for emergency responders. The proposed solution is cost-effective, scalable, energy-efficient, and suitable for deployment in forests, wildlife reserves, and remote infrastructure areas. By reducing wildfire detection time from hours to minutes, the system can significantly minimize fire damage and improve disaster response capabilities.

Keywords — IoT, ESP32, LoRaWAN, Sensor Fusion, GPS Tracking, Environmental Monitoring, Cloud Computing

I. INTRODUCTION

Wildfires are among the most destructive natural disasters, causing severe damage to forests, wildlife, ecosystems, human settlements, and the environment. The extent of damage caused by a wildfire largely depends on how quickly it is detected and controlled. Conventional wildfire detection methods, such as watchtowers, patrol teams, satellite monitoring, and public reporting, are often slow and reactive, leading to delayed responses and increased destruction. With advancements in the Internet of Things (IoT) and wireless communication technologies, it has become possible to

develop intelligent systems capable of monitoring environmental conditions in real time. This project presents an IoT-based early wildfire detection system that continuously monitors parameters such as temperature, humidity, smoke concentration, and flame presence using multiple sensors. A sensor fusion approach is employed to improve detection accuracy and minimize false alarms by analyzing data from different sensors simultaneously. The collected information is processed by a microcontroller and transmitted through long-range wireless communication networks to cloud-based platforms for monitoring and analysis. In the event of potential wildfire conditions, the system generates real-time alerts along with precise location information, enabling rapid response from emergency authorities. The proposed solution is designed to be cost-effective, energy-efficient, scalable, and suitable for deployment in remote forest regions where traditional monitoring methods are less effective. By reducing wildfire detection time from hours to minutes, the system significantly enhances disaster response capabilities, minimizes environmental and economic losses, and contributes to sustainable forest management and conservation efforts.

II. LITERATURE REVIEW

Chan et al. [1] provided a comprehensive survey on the deployment of IoT ground-sensing systems for early wildfire detection, identifying them as a critical, high-precision alternative to satellite and vision-based systems. Their research notes that while spaceborne assets (such as MODIS or Sentinel satellites) offer massive wide-area geographical

coverage, they suffer from inherent operational drawbacks: long orbital revisit cycles, cloud cover obstructions, and severe canopy occlusion that prevents the discovery of understory ignitions.

Lloret et al.[2] proposed a wireless sensor network-based forest fire detection system that continuously monitors environmental parameters such as temperature, humidity, and smoke levels. The study demonstrated that distributed sensor nodes can provide real-time monitoring over large forest areas while maintaining low power consumption. Their work highlighted the importance of early fire detection and efficient wireless communication in reducing wildfire damage and improving emergency response capabilities.

Mohamed Hefeeda and Majid Bagheri[4] proposed a forest fire detection framework using wireless sensor networks and fire-spread modeling techniques. Their work focused on optimizing sensor placement and improving detection speed while minimizing energy consumption. The study demonstrated that strategically deployed sensors can detect fires at an early stage and improve monitoring efficiency. Their research highlighted the importance of reliable communication and sensor coverage in wildfire detection systems.

Bin Son, Yong-Sik Her, and Jung-Gyu Kim[5] designed a forest fire surveillance system based on wireless sensor networks. The system continuously monitored environmental parameters and transmitted information to a central monitoring station. Their implementation proved effective for wildfire monitoring in mountainous and remote regions. The study concluded that sensor networks offer a reliable and cost-effective solution for real-time forest fire detection and environmental monitoring.

III. METHODOLOGY/EXPERIMENTAL

A. Materials

The system is built using industrial-grade non-invasive sensors and a high-performance microcontroller to ensure both safety and data accuracy.

Component	Model	Purpose
Microcontroller	ESP32 (Dual-Core)	Processes sensor data and manages communication
Temperature sensor	SHT31	Measures temperature and humidity
Gas sensor	MQ-135	Detects harmful gases such as CO ₂ and smoke
Smoke sensor	MQ-2	Detects smoke and combustible gases
Flame Sensor	IR Flame Sensor Module	Detects infrared radiation emitted by flames.
GPS Module	U-blox NEO-6M	Provides the exact location of the detected wildfire
Communication Module	LoRa SX1278 (LoRaWAN)	Enables long-range transmission of sensor data and alerts.

B. Flowchart

1. Multi-sensor arrays monitor sub-canopy thermal and chemical metrics continuously.
2. On-node ESP32 logic processes edge-level sensor fusion algorithmically.
3. NEO-6M GPS module appends precise geographic location telemetry.
4. LoRaWAN transceivers transmit data packets over 10–15 kilometers.
5. AWS backend ingests data to trigger instant emergency alerts.

C. Block Diagram

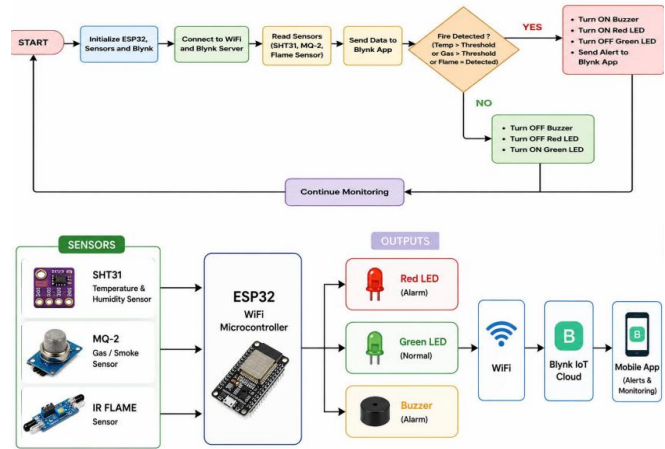


Fig.1

D. Theory of Operation

The system continuously monitors environmental conditions using temperature, humidity, smoke, gas, and flame sensors. Sensor data is analyzed to detect wildfire risks, and alerts with GPS coordinates are transmitted for rapid emergency response. The system is based on finding wildfires early. It uses Internet of Things technology and sensors to do this. These sensors are always checking things like the temperature and humidity. They also check for smoke and gas. If there are flames. The ESP32 microcontroller looks at all the information from the sensors to see if anything is not normal. This could mean a wildfire is starting. By using sensors the system is better at finding wildfires and it does not give as many false alarms. When the system thinks a fire might be starting it uses a GPS to figure out where it is.

Then it sends an alert to a cloud platform using LoRaWAN communication. The cloud platform stores the information. Looks at it. Then it sends the information to the people who deal with emergencies. This helps them respond quickly. It reduces the damage from wildfires. This way of doing things is good because it does not cost much and it is reliable. It also uses energy, which is good for continuously watching over the forest and managing disasters. The wildfire detection system is a solution, for this. The wildfire detection system helps with forest monitoring and disaster management.

IV. RESULTS AND DISCUSSIONS

The performance of this project was validated through a controlled experimental setup. The primary objective was to verify the accuracy of the sensing logic and the reliability of the cloud-integrated notification system.

A. Experimental Setup

An ESP32 microcontroller was interfaced with temperature-humidity, smoke, gas, and flame sensors. A GPS module provided location data, while LoRaWAN enabled wireless communication. The system was powered by a Li-Po battery and solar panel, and alerts were sent when fire conditions were detected

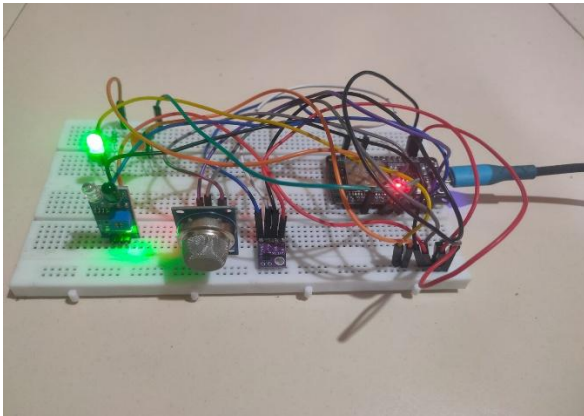


Fig.2

B. Cloud Integration

The system uses cloud integration to store and look at sensor data as it happens. It gets information from sensors that measure temperature, humidity, smoke, gas and flames. This information is sent through LoRaWAN to AWS IoT Core, where it is received and looked at in a way. The information is then put in a database called InfluxDB so we can monitor it. Look back at it later. When the system finds something that's not normal and might mean there is a wildfire it sends out alerts right away and sends them to the people who deal with emergencies along, with GPS coordinates so they can respond quickly.

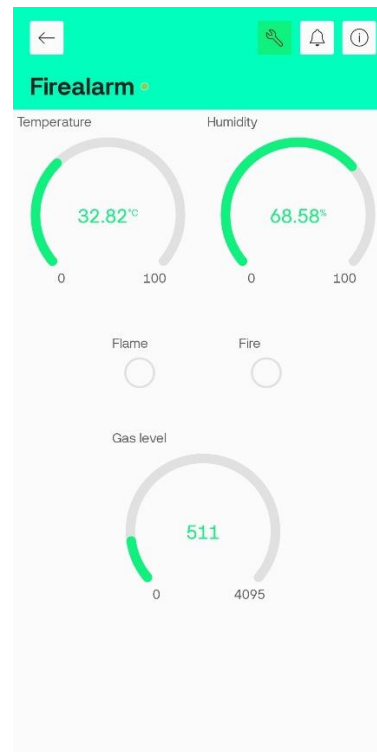


Fig.3

```
Humidity: 68.22 %
Gas Level: 551
Flame: 1
Fire Status: NO
-----
Temperature: 32.79 C
Humidity: 68.23 %
Gas Level: 557
Flame: 1
Fire Status: NO
-----
Temperature: 32.82 C
Humidity: 68.28 %
Gas Level: 552
Flame: 1
Fire Status: NO
-----
Temperature: 32.79 C
Humidity: 68.27 %
Gas Level: 551
Flame: 1
Fire Status: NO
-----
Temperature: 32.79 C
Humidity: 68.25 %
Gas Level: 544
Flame: 1
Fire Status: NO
-----
Temperature: 32.79 C
Humidity: 68.28 %
Gas Level: 546
Flame: 1
Fire Status: NO
-----
Temperature: 32.79 C
Humidity: 68.29 %
Gas Level: 549
Flame: 1
Fire Status: NO
```

Fig.4

C. Discussion

The developed wildfire detection system successfully demonstrates the use of IoT and sensor fusion for early fire detection. By continuously monitoring temperature, humidity, smoke, gas concentration, and flame presence, the system can identify potential wildfire conditions in real time. The integration of multiple sensors improves reliability and reduces false alarms compared to single-sensor systems. GPS tracking enables accurate location identification, while LoRaWAN communication ensures long-range data transmission from remote areas. Cloud integration further enhances monitoring and alert generation. The proposed system is cost-effective, energy-efficient, and suitable for

deployment in forests and other wildfire-prone regions, contributing to faster emergency response and reduced environmental damage.

V. MATH

The system continuously measures environmental parameters including temperature, humidity, smoke concentration, gas levels, and flame presence using multiple sensors. Temperature and humidity readings are obtained from the SHT31 sensor, while smoke and gas levels are monitored using MQ-2 and MQ-135 sensors. The IR flame sensor detects the presence of fire, and the NEO-6M GPS module provides the geographical coordinates of the monitored area. Sensor readings are compared with predefined threshold values, and when abnormal conditions indicating a potential wildfire are detected, an alert is generated and transmitted through the communication network for immediate action.

1. Temperature Change

$$\Delta T = T_{\text{current}} - T_{\text{normal}}$$

Where:

T_{current} = Current temperature

T_{normal} = Normal ambient temperature

2. Humidity Change

$$\Delta H = H_{\text{normal}} - H_{\text{current}}$$

Where:

H_{current} = Current humidity

H_{normal} = Normal humidity

3. Fire Risk Score

$$\text{FRS} = w_1(T) + w_2(S) + w_3(G) + w_4(F)$$

Where:

T = Temperature value

S = Smoke sensor value

G = Gas sensor value

F = Flame sensor output

w_1, w_2, w_3, w_4 = Weight factors

4. Alert Condition

"Fire Alert" = { 1, &"if " $T > T_{\text{th}}$, $S > S_{\text{th}}$, "or " }

$F = 1 @ 0, \&"otherwise" \}$

Where:

T_{th} = Temperature threshold

S_{th} = Smoke threshold

VI. UNITS

The system uses various units to measure environmental and operational parameters. Temperature is measured in degrees Celsius ($^{\circ}\text{C}$), while humidity is measured in percentage relative humidity (%RH). Smoke and gas concentrations are measured in parts per million (ppm). The GPS module provides location information in degrees (latitude and longitude). Battery performance is measured in volts (V), and the communication distance of the LoRaWAN module

is measured in kilometres (km). These units enable accurate monitoring, analysis, and transmission of wildfire-related data.

Parameter	Unit
Temperature	$^{\circ}\text{C}$ (Degree Celsius)
Humidity	% RH (Relative Humidity)
Smoke/Gas	ppm (Parts Per Million)
GPS Coordinates	km (Kilometres)
Battery Voltage	V (Volts)
Communication Range	km (Kilometres)

VII. ENVIRONMENTAL IMPACT

Wildfires cause severe environmental damage by destroying forests, vegetation, and wildlife habitats. Large-scale fires lead to the loss of biodiversity and disrupt the ecological balance of affected regions. The proposed system helps minimize these impacts by detecting fire incidents at an early stage, allowing authorities to take immediate action before the fire spreads uncontrollably. Forest fires release significant amounts of carbon dioxide, carbon monoxide, and other harmful gases into the atmosphere, contributing to air pollution and climate change. By reducing the size and duration of wildfires, the system helps lower greenhouse gas emissions and improves air quality in nearby communities. This contributes to a healthier environment and supports global efforts to combat climate change. The protection of forests also plays a vital role in conserving natural resources such as soil, water, and vegetation. Early wildfire detection prevents excessive damage to these resources, reducing soil erosion and preserving watersheds that are essential for maintaining healthy ecosystems. As a result, the system supports long-term environmental sustainability and ecosystem conservation. In addition, the use of solar-powered sensor nodes makes the system energy-efficient and environmentally friendly. The reliance on renewable energy reduces the need for external power sources and enables sustainable operation in remote forest areas. Thus, the project not only helps prevent environmental destruction but also promotes the use of green technology for disaster management and environmental protection.

VIII. FUTURE SCOPE

The current project can be made better by adding sensor nodes. This will help cover more of the forest and make detection more reliable. We can also use techniques to combine data from sensors. This will help reduce alarms and make wildfire detection more accurate in different weather conditions. We can make the communication network bigger by adding LoRaWAN gateways. This will allow us to

monitor forests that're really far away. The wildfire detection system can also be connected to web applications. This will provide real-time alerts and live monitoring. It will also be easier for forest officials and emergency response teams to get to the sensor data they need. In the future we might add intelligence and machine learning to the wildfire detection system. This will help us look at trends, in the environment. Predict when wildfires might happen before they start. We can also use drones and satellite images to verify what the sensors detect. This will help us manage disasters better. We can also make the system better by using power efficiently and adding bigger solar panels. This will help the wildfire detection system work on its own for a time. These changes will make the wildfire detection project work better cover area and be more practical to use. The wildfire detection project will be more efficient and scalable. The wildfire detection project will be better overall.

IX. CONCLUSION

This automated, low-cost early wildfire detection network successfully shifts the forest monitoring paradigm from slow, reactive human observations to a proactive, 24/7 intelligent ground-level framework. By deploying optimized edge nodes utilizing an ESP32 microcontroller, SHT31 micro-sensors, MQ-135/MQ-2 gas arrays, and IR flame sensors directly within high-risk understories, the system calculates a multi-modal "sensor fusion" index that effectively eliminates false alarms and detects ignitions within minutes rather than hours. Once a localized fire is verified at the edge, a U-blox NEO-6M GPS module logs its precise coordinates, and a long-range LoRaWAN transceiver transmits the data packets up to 10–15 km to regional elevated gateways for instant AWS cloud ingestion, time-series logging, and automated emergency notification. In summary, by utilizing a rugged IP67-rated enclosure and a standalone 3.7V LiPo battery paired with a 6V solar panel, this network operates as a robust, scalable, and economically feasible "deploy and forget" infrastructure that offers a multi-year solution to a multi-billion dollar global environmental problem. Furthermore, the operational versatility of this distributed grid introduces a scalable foundation for safeguarding critical infrastructure, wildland-urban interfaces, and state forestry assets from escalating climate threats. Beyond rapid suppression alerts, the continuous time-series environmental data captured by the network provides researchers with high-resolution micro-climatic datasets, while modular firmware hooks allow for seamless future integrations of soil moisture and geological sensors for secondary landslide or flood tracking. By achieving localized intelligence directly at the edge, the system successfully bridges the critical visibility gap left by macro-scale satellite remote sensing and visual watchtowers. Ultimately, this architecture establishes that a dense network of highly affordable, ultra-low-power IoT nodes can deliver the immediate, target-specific detection capabilities required to neutralize localized ignitions before they evolve into catastrophic, unmanageable disasters

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REFERENCES

- [1] J. Lloret, M. Garcia, D. Bri, and S. Sendra, "A Wireless Sensor Network Deployment for Rural and Forest Fire Detection and Verification," *Sensors*, vol. 9, no. 11, pp. 8722–8747, 2009.
- [2] C. Hartung, R. Han, C. Seielstad, and S. Holbrook, "FireWxNet: A Multi-Tiered Portable Wireless System for Monitoring Weather Conditions in Wildland Fire Environments," *Proceedings of MobiSys 2006*, pp. 28–41, 2006.
- [3] M. Hefeeda and M. Bagheri, "Forest Fire Modeling and Early Detection Using Wireless Sensor Networks," *Ad Hoc & Sensor Wireless Networks*, vol. 7, pp. 169–224, 2009.
- [4] B. Son, Y. Her, and J. Kim, "A Design and Implementation of Forest-Fires Surveillance System Based on Wireless Sensor Networks for South Korea Mountains," *IJCSNS International Journal of Computer Science and Network Security*, vol. 6, no. 9, pp. 124–130, 2006.
- [5] K. Sha, W. Shi, and O. Watkins, "Using Wireless Sensor Networks for Fire Rescue Applications: Requirements and Challenges," *IEEE International Conference on Electro/Information Technology*, pp. 239–244, 2006.
- [6] D. M. Doolin and N. Sitar, "Wireless Sensors for Wildfire Monitoring," *Smart Structures and Materials: Sensors and Smart Structures Technologies for Civil, Mechanical, and Aerospace Systems*, 2005.
- [7] J. Toledo-Castro, I. Santos-González, P. Caballero-Gil, C. Hernández-Goya, N. Rodríguez-Pérez, and R. Aguasca-Colomo, "Fuzzy-Based Forest Fire Prevention and Detection by Wireless Sensor Networks," 2022.
- [8] K. K. Pandey and A. K. Gupta, "On Detection of Critical Events in a Finite Forest Using Randomly Deployed Wireless Sensors," 2019.